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IN ASTROPHYSICS AND SPACE SCIENCE

MASTER'S THESIS

RECONSTRUCTING PHOTOMETRIC LIGHTCURVES
OF ACTIVE GALACTIC NUCLEI USING NEURAL
STOCHASTIC FLOWS

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УНИВЕРЗИТЕТ У БЕОГРАДУ
МАТЕМАТИЧКИ ФАКУЛТЕТ



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Реконструкција фотометријских светлосних
кривих активних галактичких језгара
применом стохастичких Neural Flow модела

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Contents

1	Background	7
1.1	Unified Model of AGN	9
1.1.1	Accretion Disk	11
1.1.2	Corona	12
1.1.3	Broad-line Region	12
1.1.4	Dusty Torus	13
1.1.5	Relativistic Jets	14
1.1.6	Narrow-line Region	15
2	Variability of AGN	16
2.1	Physical Origin and Timescales	16
2.1.1	Reverberation Mapping	18
2.2	Variability Characterization	19
3	Simulation Framework	21
3.1	Stochastic Processes	21
3.1.1	The Transition Probability	22
3.2	The Damping Timescale	23
3.3	Physically Motivated Timescales	24
3.4	Light Curve Generation	24
3.5	Sampling Regime	26
4	Neural Stochastic Flows	27
4.1	Normalizing Flows	27
4.2	From Static Densities to Transition Laws	28
4.3	Triplets and Stochastic Bridges	30
4.4	Training Objective	31
4.5	Implementation	31

5	Results and Discussion	33
5.1	Fixed Characteristic Timescale τ	33
5.2	Sampling timescales	33
	Bibliography	34

List of Figures

1.1	AGN spectral energy distribution (SED) from radio frequencies to hard X-ray . .	8
1.2	Typical optical spectra of different AGN classes	9
1.3	NGC 1068 spectropolarimetry revealing the hidden broad line region (BLR) . . .	10
1.4	Schematic view of the the innermost regions of (non-jetted) AGN	11
1.5	Eddington ratio distribution of broad-line AGN at $1.0 < z < 2.2$	12
1.6	Schematic of the BLR formation via a failed disk wind	13
1.7	Radio contours of a giant radio galaxy extending to Mpc scales	14
1.8	SDSS galaxies in the classical BPT diagram.	15
2.1	Continuum and reverberation lags as a function of energy and Fourier frequency	17
2.2	Schematic of AGN reverberation mapping across different physical scales	18
2.3	X-ray light curves and PSDs for four AGN spanning four decades in black hole mass	20
3.1	The structure-function family and the place of the damped random walk	23
3.2	The black hole mass distribution and the induced damping timescale distribution	24
3.3	Example simulated light curves	25
3.4	Sampling regime of the simulated light curves	26
4.1	The change of variables underlying a normalizing flow	28
4.2	Strong and weak solutions of an SDE and one-step sampling	29

List of Tables

2.1	AGN region size scales and variability timescales	18
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1 Background

Compared to ordinary galaxies, some have extraordinarily bright central regions, which can sometimes even outshine the host-galaxy emission by many orders of magnitude; these galaxies are called *active galaxies* and their nuclei are referred to as *active galactic nuclei* (AGN). The physical processes necessary to create such high luminosities ($L = 10^{43-48} \text{ erg s}^{-1}$; e.g., Lusso et al. 2012) within such a small volume were a subject of debate for years, but now are widely accepted to be connected to accretion onto the galaxies' central black holes (BHs; Salpeter 1964; Zeldovich 1964). These accreting objects have masses $M_{BH} \sim 10^{6-10} M_{\odot}$ (e.g., Peterson and Horne 2004) and an Eddington ratio exceeding $\lambda_{Edd}^1 \approx 10^{-5}$. This threshold, chosen somewhat arbitrarily, excludes galactic nuclei with no significant accretion activity, such as that of the Milky Way, while still covering their full range of observed luminosities (Netzer 2015).

The “unified model” (Antonucci 1993; Urry and Padovani 1995) discussed in Section 1.1, posits the black hole at the center of the AGN, surrounded by several distinct components, including an accretion disk, gas and dust clouds, and, in some cases, a relativistic jet (Figure 1.4). Each of these regions contributes emission into different wavelength regimes, resulting in these objects having detectable emission across the whole electromagnetic spectrum, as illustrated by the spectral energy distribution (SED) in Figure 1.1.

The gravitational energy released as the circumnuclear material falls inwards and loses angular momentum forms the accretion flow around the black hole, which acts as the primary source of AGN continuum emission. The disk gets heated up via viscous heating (Peterson 1993), producing a thermal spectrum that has a blackbody temperature peaking in the ultra violet (UV; $\approx 10 - 400 \text{ nm}$) for a “typical” AGN (i.e., with $M_{BH} \approx 10^8 M_{\odot}$; $\lambda_{Edd} \approx 0.1$); this is often referred to as the “big blue bump” (Harrison 2014; Figure 1.1)

The thermal disk photons, also called *seed photons*, are Comptonized by highly energetic electrons that are located at the *corona* (§1.1.2) to form the X-ray power-law spectrum observed in Figure 1.1. These high energy photons can also reflect off the accretion disk and/or the *torus* (§1.1.4), producing the additional hard X-ray “reflection” component. In a majority of AGN X-ray spectra, there is an additional “soft excess” (Singh et al. 1985; Arnaud et al. 1985) at energies less than $\sim 2 \text{ keV}$. This emission is above what would be expected from simple accretion disk models and its origin is still a topic of discussion. Two proposed models (which could be simultaneously true) to explain this soft excess are: relativistically blurred ionized reflection from the

¹The ratio L/L_{Edd} , where L is the bolometric luminosity and $L_{Edd} = 1.5 \times 10^{38} M_{BH}/M_{\odot} \text{ erg s}^{-1}$ is the Eddington luminosity for a solar composition gas. This is the maximum isotropic luminosity a body can achieve while maintaining balance between radiation pressure and gravitational force.

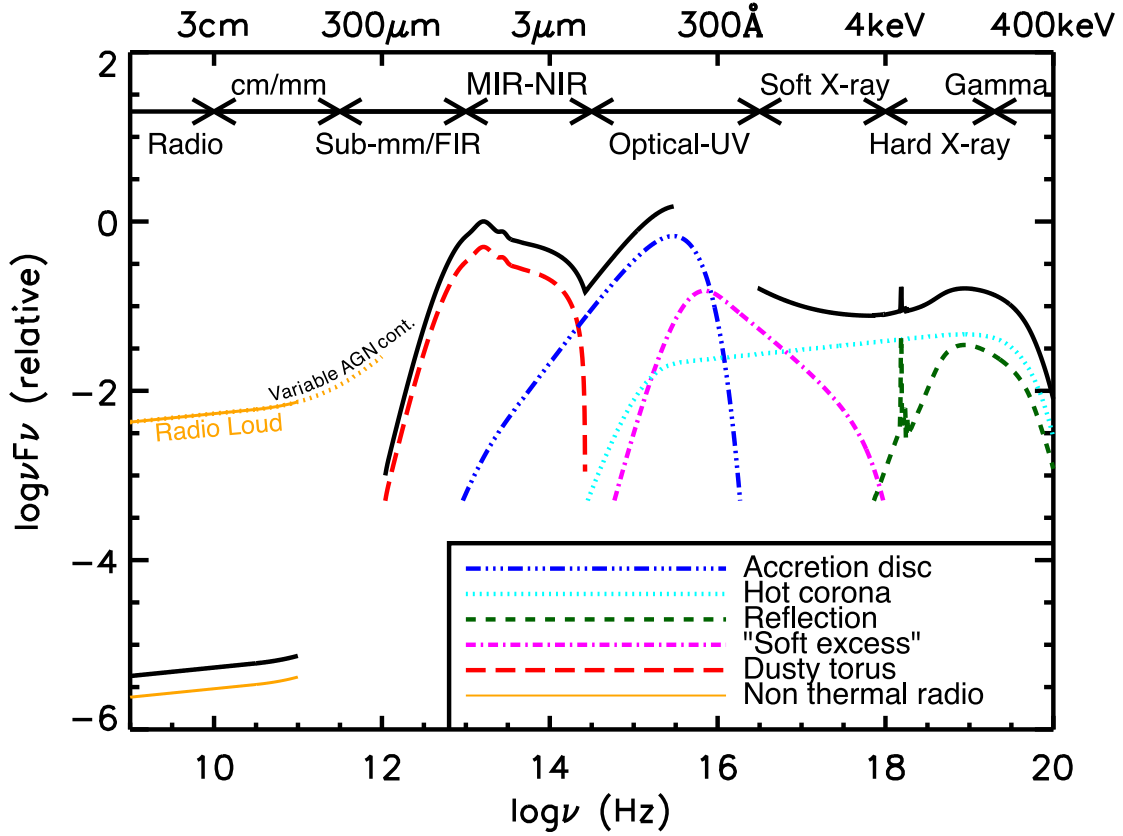


Figure 1.1: AGN SED from radio frequencies to hard X-ray, loosely based on the observed SEDs of radio-quiet quasars (e.g., Elvis et al. 1994; Richards et al. 2006). The black solid curve represents the total emission and the colored curves (shifted down for clarity) represent the individual constituents. The intrinsic shape of the SED in the mm-far infrared (FIR) regime is uncertain; however, it is credited to have a minimal contribution (to an overall galaxy SED) compared to star formation (SF), except in the most intrinsically luminous quasars and powerful jetted AGN. It should be noted that the relative contribution of each of the components can vary dramatically for different AGN types. Adapted from Harrison (2014).

inner accretion disc (e.g., Crummy et al. 2006), or Comptonization in a “warm corona” located near the surface of the accretion flow (e.g., Ballantyne and Xiang 2022).

The infrared (IR; $\approx 1 - 1000 \mu\text{m}$) band, is mostly sensitive to the reprocessed emission of the accretion disk by the torus. Although the shape of this IR emission varies from source to source, the peak of this dusty component is usually around $\lambda \approx 20 - 50 \mu\text{m}$ and falls steeply at longer wavelengths (Harrison 2014). In the sub-mm/far-infrared (FIR), or sometimes even in the mid-infrared (MIR), stellar heated emission from cooler dust dominates over AGN heated emission (Lutz 2014).

AGN cover a wide range of radio luminosities, which can differ by several orders of magnitude between radio-quiet (RQ) and radio-loud (RL) sources, now more aptly referred to as non-jetted and jetted, respectively ² (orange lines in Figure 1.1). The latter type makes up only a minority of the AGN population (e.g., Padovani 2011) with objects that present highly collimated, relativistic radio *jets* (§1.1.5) and extended radio structures such as lobes and hotspots, that together emit strong non-thermal radiation and, occasionally, a γ -ray jet. On the other hand, the spectrum of non-jetted objects is dominated by thermal emission in most bands and the origin

²While RL/RQ has been the historical terminology, radio-quiet sources are not truly radio-silent but rather “radio-faint” and they are, however, γ -ray quiet (Padovani 2017).

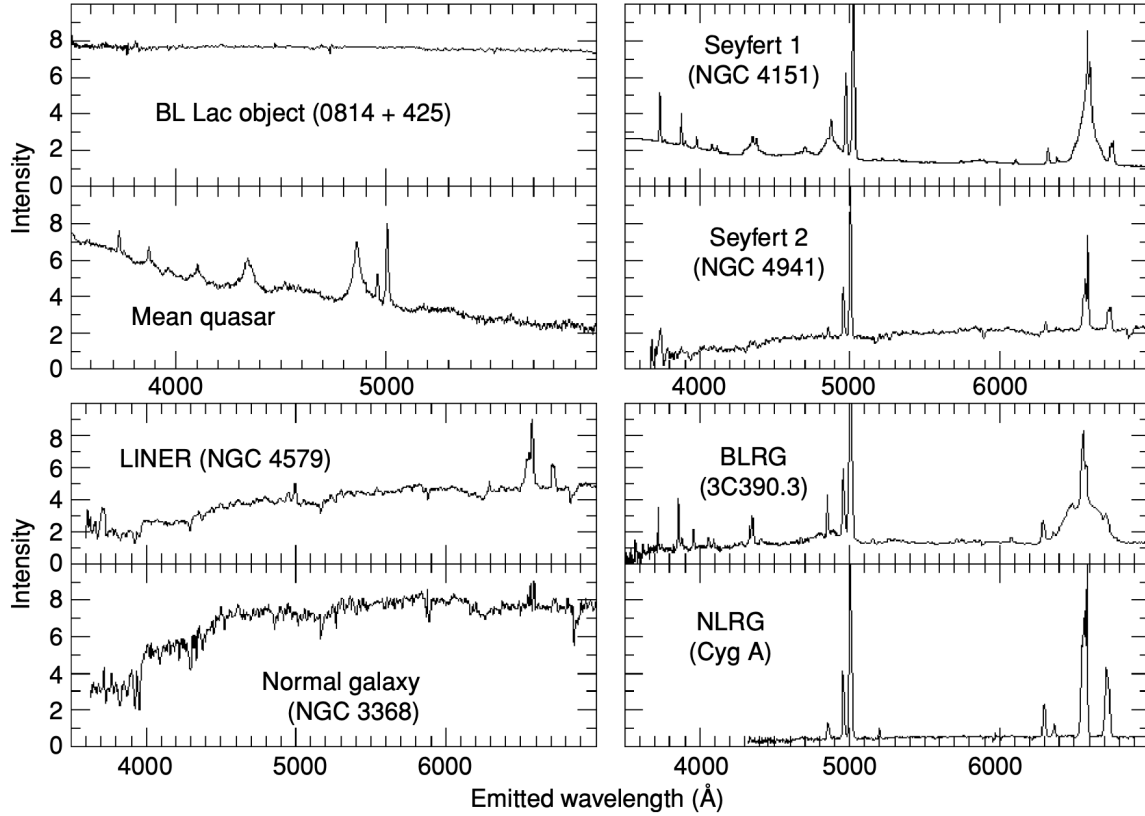


Figure 1.2: Optical spectra of representative AGN classes, from the (essentially featureless) BL Lac objects (Top left) through the radio galaxies and QSOs (Seward and Charles 2010). Note the presence of both broad and narrow emission lines in quasars, Seyfert 1 galaxies and broad-line radio galaxies (BLRG), while Seyfert 2 galaxies, low-ionisation nuclear emission region (LINER) galaxies and narrow-line radio galaxies (NLRG) only show narrow permitted and forbidden lines. A normal galaxy spectrum (Bottom left) is shown for comparison.

of their radio emission has not been clearly established. Some possibilities include coronal activity or synchrotron radiation from particles accelerated in shocks (see e.g., Laor and Behar 2008; Ishibashi and Courvoisier 2011).

1.1 Unified Model of AGN

Because different wavebands probe different physical elements, AGN selected at particular wavelengths often exhibit distinct properties, leading to a rich phenomenological classification (Figure 1.2). Reality, however, is much simpler because now we know that most of these classes differ from each other only by a small number of parameters, such as: orientation, accretion rate, the presence of jets, and possibly the host galaxy and environment (see Padovani et al. (2017) and references there-in). This suggests that the numerous brands do not necessarily correspond to fundamentally different physical objects, but instead motivate a unifying framework capable of explaining their apparent diversity.

This unification idea arose shortly after the discovery of *quasars* (QSOs³), when the observed

³The term quasar historically stands for “quasi-stellar radio source”; throughout this thesis it is used more broadly to refer to very luminous AGN.

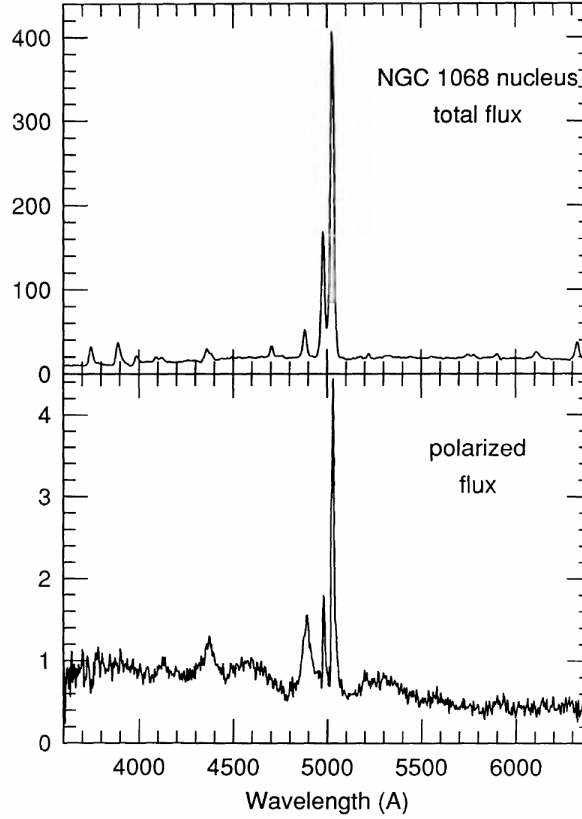


Figure 1.3: Optical spectra of the nucleus of NGC 1068 (Miller, Goodrich, and Mathews 1991). The total spectrum (Top) shows the narrow emission lines expected for Type 2 AGN, while the polarized flux spectrum (Bottom) reveals the broad permitted emission lines characteristic of unobscured Type 1 AGN.

variety of AGN prompted questions about the role of the observer’s orientation relative to the AGN axis. Blandford and Rees (1978) proposed that BL Lac objects were radio galaxies viewed down the axis of a relativistic jet, where *relativistic beaming* (§1.1.5) would cause the non-thermal continuum to appear very bright and the emission lines weak in comparison. The same object, viewed from the side, would have normal emission-line equivalent widths, and the radio structure would be dominated by the extended lobes rather than the core (Shields 2004).

Further evidence came from spectropolarimetry, the measurement of the polarization state of the incoming light from a source as a function of wavelength. Seyfert galaxies (Seyfert 1943) are primarily divided into two types based on their optical spectra: Type 1 objects exhibit both broad permitted emission lines such as $H\alpha$ and $H\beta$ ($\text{FWHM} > 1000 \text{ km s}^{-1}$) and narrow forbidden lines such as $[\text{O III}]$ and $[\text{Ne III}]$ ($\text{FWHM} < 1000 \text{ km s}^{-1}$), while Type 2 objects show only the narrow component. Crucially, both types share the same high- and low-ionization narrow forbidden lines with similar line ratios, suggesting they are powered by the same intrinsic engine.

Antonucci and Miller (1985) confirmed this by analyzing the polarized flux of NGC 1068, a Seyfert 2 galaxy⁴, resembled that of a Seyfert 1 spectrum, as can be seen in Figure 1.3. This was interpreted as the broad-line region (BLR; §1.1.3) and the central continuum source being blocked from direct view by a dusty torus, while free electrons above the nucleus scatter a fraction of the nuclear light towards the observer, polarizing it in the process. The broad lines had gone unnoticed in direct light because this scattered component was too faint relative to the

⁴It should be noted that a subset of Type 2 AGN ($\sim 30\%$), sometimes referred to as “true Type 2” objects, show no detectable broad lines even in polarized light and little X-ray absorption (see Netzer 2015 and references there-in).

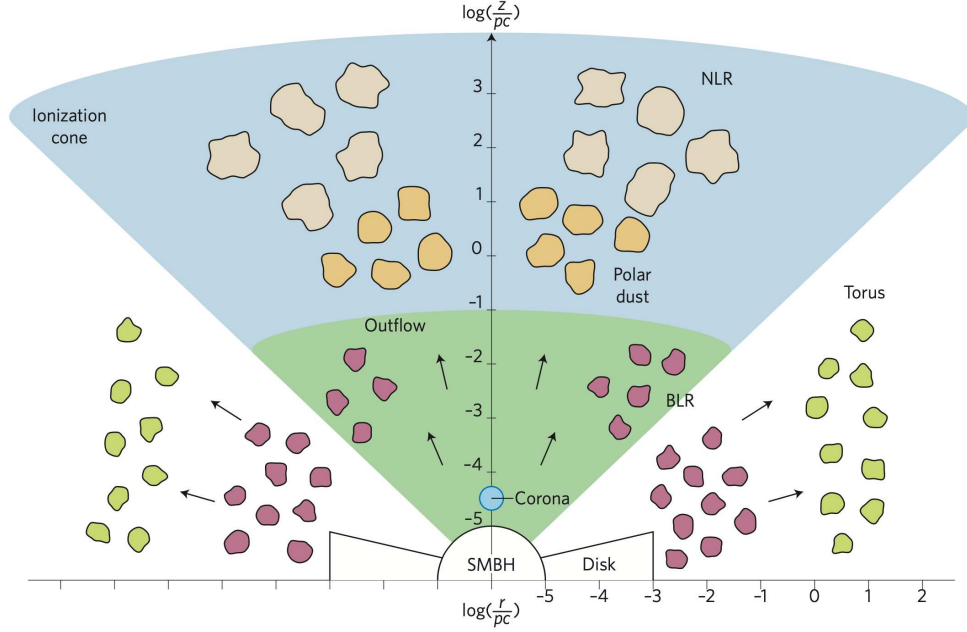


Figure 1.4: Overview of the innermost regions around a non-jetted accreting supermassive black hole (SMBH). Typical scales along the polar and equatorial directions (for nearby AGN) are shown. Different colors depict regions with different densities or compositions. Taken from Ramos Almeida and Ricci 2017.

strong narrow-line emission from the narrow-line region (NLR; §1.1.6), which extends beyond the obscuring torus and is therefore visible regardless of orientation.

Multiple efforts such as those recounted above (e.g., Antonucci 1984; Orr and Browne 1982) were fundamental to the development of the “favoured” unified model of AGN introduced in this section. Whilst it has been successful in homogenizing many kinds of active nuclei, recent works point to the conclusion that a picture based primarily on orientation and obscuring material is incomplete (e.g., Bianchi, Maiolino, and Risaliti 2012). Nonetheless, the major aspects of the model are sufficient for the purposes of this thesis; in the sections below, the primary components are explained briefly, while additional features such as polar dust and large-scale outflows (also visible in Figure 1.4) are discussed shortly when relevant but are not treated as separate subsections.

1.1.1 Accretion Disk

The standard framework of the optically thick, geometrically thin accretion disk (Shakura and Sunyaev 1973) described above applies to SMBHs with moderate to high Eddington ratios ($10^{-2} \lesssim \lambda_{Edd} \lesssim 1$; Ricci 2026), where the accretion flow is considered radiatively efficient. This scheme describes the disk as a composition of Keplerian rings, nested and coupled to each other by viscous forces, likely driven by turbulence and magnetic instabilities, the magneto-rotational instability (MRI; Balbus and Hawley 1998) being the most widely quoted. The outer, slower ring brakes the inner faster one, transporting the angular momentum outwards and allowing material to spiral inward. Friction between the rings produces heat that is dissipated as blackbody emission with a characteristic maximum temperature of $T \sim 10^5$ K for the “typical” AGN described above (Netzer 2013), yielding the multi-temperature spectrum depicted in Figure 1.1. The continuum emission of the accretion disk varies with time depending on the dynamics of the infalling matter (see Chapter 2).

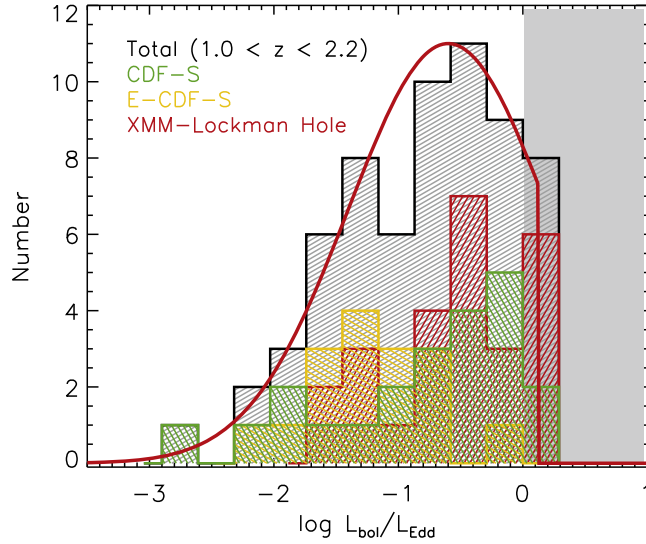


Figure 1.5: Eddington ratio distribution of broad-line AGNs at $1.0 < z < 2.2$. Different X-ray surveys are represented as different color histograms, and the black histogram pictures all the combined distributions. The gray shade illustrates the Eddington limit ($\lambda_{Edd} = 1$) and the red solid line corresponds to a log-normal fit with a peak of $\log L/L_{Edd} = -0.6$ (Suh et al. 2015).

Most QSOs in the local universe are thought to accrete through thin disks, yet contrasting accretion scenarios are possible depending on the Eddington ratio (Figure 1.5). At $\lambda_{Edd} \lesssim 10^{-2}$, the disk becomes optically thin and with an Advection Dominated Accretion Flow (ADAF, Narayan and Yi 1994), radiating weakly and with minimal optical/UV emission. Conversely, at $\lambda_{Edd} \gtrsim 1$, theoretical works have predicted that the SMBHs accrete rapidly through “slim” (or “thick”) disks, showing very different SEDs than their slower counterparts and powerful outflows (e.g., Kubota and Done 2018; Giustini and Proga 2019).

1.1.2 Corona

Numerical simulations of magnetohydrodynamical accretion disks have shown that the dissipation of the magnetic energy built up by the MRI naturally produces a hot, active corona located within a few gravitational radii ($R_g = GM/c^2$) of the accretion disk (Balbus and Hawley 1998; Miller and Stone 2000; Fabian et al. 2009). This compact region of ionized gas reaches temperatures of $T \sim 10^9$ K (Kara and García 2025), and its rapid variability (see Chapter 2) is consistent with such a small emitting region. The corona contributes approximately 5% of the AGN bolometric output (Ricci 2026), a fraction that decreases with increasing λ_{Edd} , with lower λ_{Edd} systems showing stronger X-ray emission relative to the disk (Padovani et al. 2017).

1.1.3 Broad-line Region

The broad-line region consists of high density gas clouds ($\gtrsim 10^9 \text{ cm}^{-3}$) predominantly in Keplerian motion at luminosity-dependent distances of $0.01 - 1$ pc from the central black hole (Netzer 2015), well within the *dust sublimation radius* (see §1.1.4). The BLR is therefore dust free, emitting broad permitted and semi-forbidden emission lines with velocity widths ranging from 1,000 to 20,000 km s^{-1} (Ricci 2026), wideness caused by *Doppler broadening*⁵. Ionizing photons from the

⁵Doppler broadening refers to the superposition of emission from clouds moving at different directions/velocities along the line of sight produces a single broadened line whose width reflects the velocity dispersion of the gas.

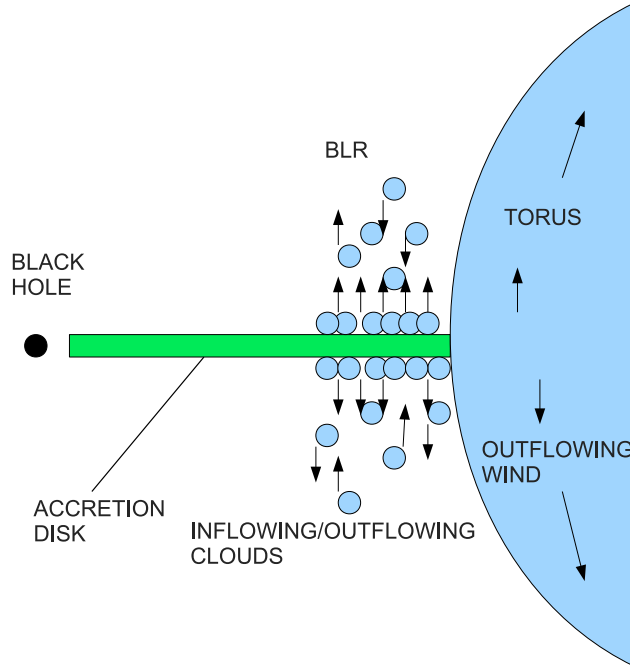


Figure 1.6: Schematic of the BLR geometry and its location relative to the accretion disk and dusty torus. The figure also illustrates the failed wind scenario of Czerny and Hryniewicz (2011): dust forms in the atmosphere of the inner accretion disk at radii where $T_{eff} < T_{sub}$, is launched as a wind, but then is suppressed by the radiation from the central source and partially falls back, giving rise to the inflowing and outflowing clouds that make up the BLR.

accretion disk and corona illuminate these clouds, which subsequently exhibit ionization stratification with respect to radius, with higher ionization species located closer to the central source (Netzer 2013). The physical origin of the BLR remains an open question, some of the models presented in the literature include radiation pressure driven disk winds (Murray et al. 1995) and failed dust-driven wind from the disk surface (Figure 1.6; Czerny and Hryniewicz 2011). Notably, the distance of the BLR from the central source scales with the AGN luminosity as $R_{BLR} \propto L^{1/2}$ (Kaspi et al. 2000), a relationship that is fundamental in reverberation mapping studies (§2.1.1).

1.1.4 Dusty Torus

As posed previously, the presence of an assymetric “dusty obscurer” has successfully unified apparently different AGN classes. When observed edge-on (i.e., at high inclination angles), this torus⁶ obscures the innermost regions of the AGN and hides the broad emission lines; conversely, when viewed face-on (at lower inclination angles), there is a clear line of sight to the BLR and accretion disk (Seyfert 1 vs Seyfert 2 in Figure 1.2). This structure is conceptualized as an optically thick structure surrounding the central engine at scales larger than that of the BLR and luminosity dependent dimensions ranging from 0.1 – 10 pc (Netzer 2015). Its inner boundary coincides is the dust sublimation radius (R_{sub}), beyond which the dust can survive the radiation from the central engine. This radius corresponds to the distance at which the dust temperature reaches the sublimation threshold ($T_{sub} \sim 1400 - 1800$ K for silicate and graphite grains respectively; e.g., Barvainis 1987; Netzer 2015).

⁶For convenience, we keep referring to this obscuring matter as the torus despite the fact that it could have a different (and more complex) geometrical distribution (e.g., Nenkova et al. 2008; Stalevski et al. 2012).

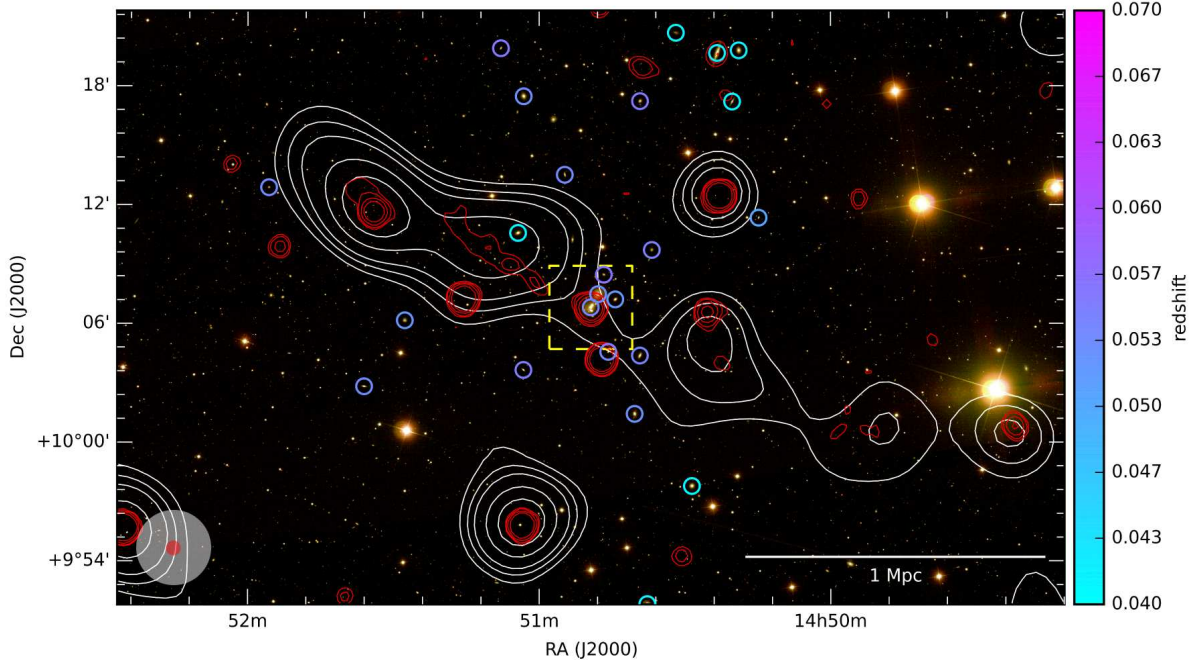


Figure 1.7: SDSS optical image (composite from bands g , r and i) of the galaxy group UGC 9555 overlaid with NVSS contours (red) and LOFAR M8SS radio contours (white) revealing the size of the giant radio galaxy (GRG). The white bar indicates a physical scale of 1 Mpc, illustrating how AGN jets can extend well beyond the boundaries of their host galaxies. Adapted from Clarke et al. 2017.

In addition to being responsible for the IR reprocessed emission on the AGN SED (Figure 1.1), the torus acts as a gas and dust reservoir that regulates the material available for accretion onto the black hole. Its geometry, believed to contain a central opening, also collimates the AGN radiation into bi-conical ionization cones along the polar direction (Pogge 1988), which illuminate the NLR (§1.1.6). Furthermore, AGN drive large scale winds and outflows along these directions, powered by radiation and magnetic pressure from the central engine, which can interact with and regulate the host galaxy interstellar medium (Ricci 2026).

1.1.5 Relativistic Jets

Jets are thought to be powered by the extraction of rotational energy from the SMBH via strong electromagnetic fields (Blandford and Znajek 1977), and can extend to Mpc scales from their host galaxy (Figure 1.7; Clarke et al. 2017).

Their launch point coincides with the location of the corona, in the immediate vicinity of the accreting object, and their apparent velocities range from mildly to highly relativistic motion. The broadband emission of jets is characterised by a double-peaked SED: the low-energy peak, spanning radio to optical wavelengths, arises from synchrotron radiation of relativistic electrons spiralling in the jet magnetic field, while the high-energy peak, from X-ray to γ -ray energies, is thought to result from inverse-Compton scattering of soft photons from the BLR and accretion disk by the same relativistic electrons (Netzer 2013).

Their observed properties depend strongly on orientation because relativistic beaming could make an approaching jet appear significantly brighter than a receding one. This effect, known as *Doppler boosting*, arises when the jet plasma has bulk relativistic motion that concentrates the emitted radiation into a narrow cone in the direction of motion, amplifying the observed flux and variability for jets viewed close to face-on and explaining the extreme characteristics observed

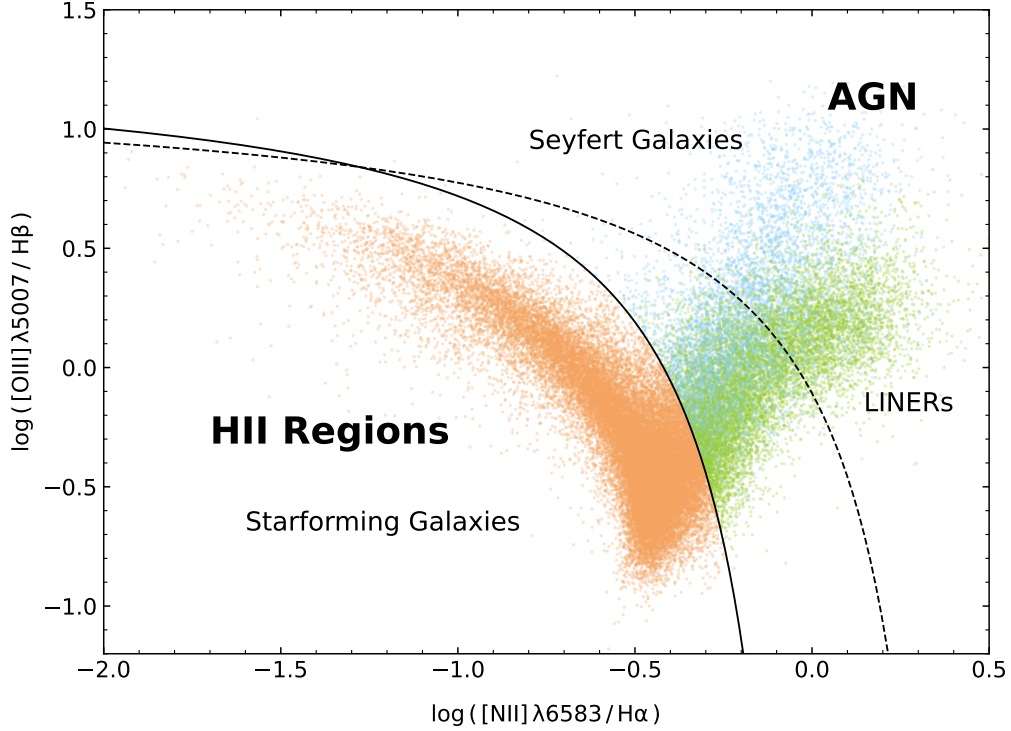


Figure 1.8: Select galaxy objects from the Sloan Digital Sky Survey (SDSS) catalog shown in the BPT diagram. It plots the intensity of the [OIII] $\lambda 5007$ forbidden transition relative to that of the $H\beta$ line versus the ratio of [NII] $\lambda 6583$ to $H\alpha$ on logarithmic axes. Using this to infer ionization states and abundances of the emitting medium enables us to reconstruct the origin of the dominant radiation in the galactic host, which has led to various approaches for the classification of these objects. Depicted is the theoretical delimiter by Kewley et al. (2001) as well as the empirical line by Kauffmann et al. (2003) to distinguish predominantly starforming galaxies from AGN. Additionally, the Schawinski et al. (2007) boundary sorts AGN into efficiently accreting Seyfert and inefficient LINER subtypes. Courtesy of Fritz A. Agildere.

on blazars (Urry and Padovani 1995). Jets are also thought to play a role in galaxy evolution by injecting energy into the surrounding environment, a process referred to as radio-mode feedback (Fabian 2012).

1.1.6 Narrow-line Region

This assymetric region is extended enough to be resolved optically (e.g., Fischer et al. 2013; Polack et al. 2024), spanning hundreds or even thousands parsecs along the ionization cones (Figure 1.4). The NLR, just as the BLR, is photoionized by the central engine and exhibits optical emission lines, kinematically broadened with typical widths of $10^2\text{--}3 \text{ km s}^{-1}$ (Bennert 2005), with most lines falling within the $350 - 500 \text{ km s}^{-1}$ range (Ricci 2026).

Compared to the BLR, the NLR has a lower electron density ($\sim 10^{2-3} \text{ cm}^{-3}$), allowing forbidden transitions to occur without being collisionally suppressed. The presence of these forbidden lines, together with the high ionization state of the gas, can trace AGN activity and makes the NLR emission diagnostically distinct from that of star-forming regions, a separation commonly visualised using emission line ratio diagrams such as the Baldwin-Phillips-Terlevich (BPT) diagram (Baldwin, Phillips, and Terlevich 1981) (Figure 1.8).

2 Variability of AGN

The first quasar discovered, 3C 273 (Edge et al. 1959), had been documented in photographic plates at least as early as 1887 (Slavcheva-Mihova et al. 2013). This emblematic radio-loud object showed variable emission in the optical and radio bands on timescales from days to years (Smith and Hoffleit 1963; Dent 1965), indicating that the size of the emitting region spanned only a few light-days in size. This compactness, coupled with the finding of its distant redshift (Schmidt 1963), was very difficult to explain by dynamics that operate in normal galaxies, and variability thus became one of the key arguments that the enormous luminosities of quasars had to originate from an exceptionally compact and energetic central engine (Salpeter 1964; Zeldovich 1964).

Now we know that variability in all wavelengths is an ubiquitous property of AGN, and so, its studies have been essential in understanding the physics of their nuclear regions, which, in general, cannot be resolved by current telescopes. This had been strictly true only until the observations of Sgr A* and M87 (The Event Horizon Telescope Collaboration et al. 2019) and, more recently, with the GRAVITY interferometer on the VLT resolving the outer BLR in 3C 273 (Gravity Collaboration et al. 2018); still, these surveys are limited to massive sources in the very nearby universe. For the vast majority of AGN, variability studies therefore rely on the *light curve*, a time series recording how the brightness (or flux) of the AGN, measured within a given wavelength band, varies over a monitoring period. The time scales, spectral changes, and the correlations and delays between variations extracted from these light curves provide crucial information on the nature and location of these components and on their interdependencies (Ulrich et al. 1997).

While variability-based methods are, generally speaking, more effective than other methods (e.g., SED-based) at providing a clearer signature of the nuclear activity against host galaxy emission (Yu et al. 2025), deriving the features and their uncertainties from an irregularly sampled light curve is very challenging; thereon, high quality light curves are indispensable to enable its robust characterization (Kelly et al. 2014).

2.1 Physical Origin and Timescales

Early campaigns quickly revealed that AGN vary much more rapidly in X-rays than in the optical (Winkler and White 1975), sometimes on timescales as short as a few minutes (McHardy et al. 1981), and that variability at different wavelengths is correlated (Ulrich et al. 1997). This stochastic behaviour can generally be explained by the *propagating fluctuations* theory (Lyubarskii 1997).

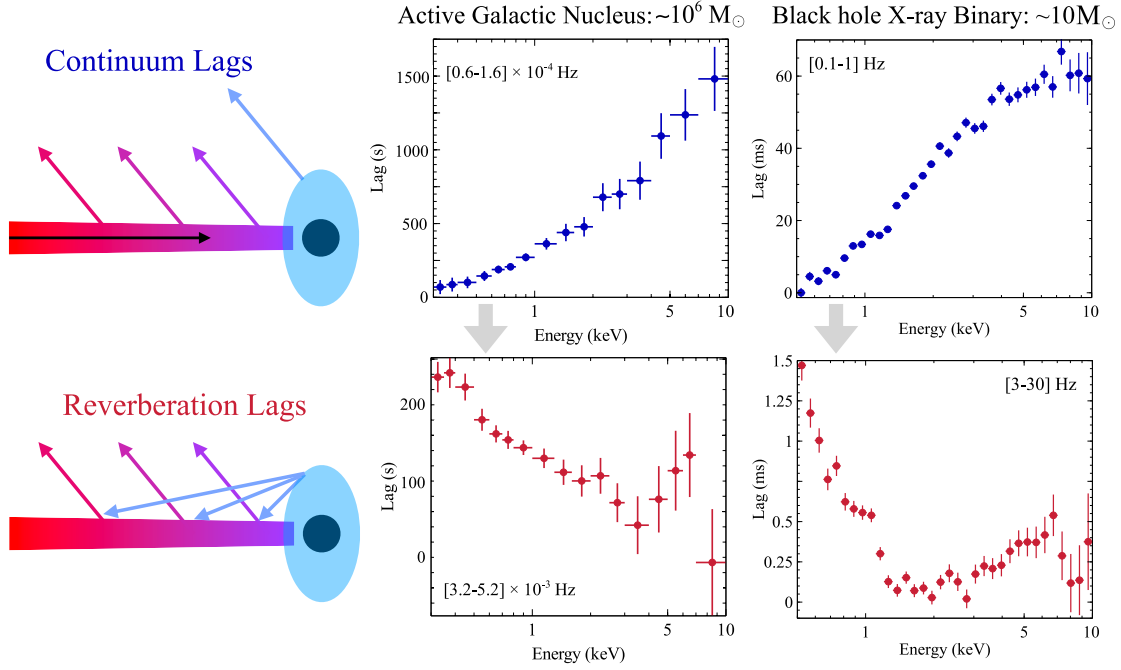


Figure 2.1: Frequency-resolved X-ray time lags as a function of energy for an AGN ($\sim 10^6 M_\odot$, left) and a black hole X-ray binary ($\sim 10 M_\odot$, right), illustrating the universality of the two lag regimes. At low Fourier frequencies (top, blue), hard X-rays lag soft ones, consistent with inward-propagating accretion fluctuations. At high Fourier frequencies (bottom, red), soft X-rays lag hard ones, consistent with disk reverberation. Taken from Kara and García 2025.

Local fluctuations in the disk’s surface density, presumably driven by the MRI (§ 1.1.1), occur at all radii and propagate inwards at the local viscous timescale¹. As these fluctuations travel, they modulate the local mass accretion rate at progressively smaller radii, coupling the variability of the outer disk to that of the hotter inner regions and corona, where the shorter local viscous timescale drives faster fluctuations in the emitted luminosity (Uttley et al. 2023). The X-ray variability produced in the innermost regions (i.e., in the corona) then irradiates the outer accretion disk and is reprocessed, driving the optical and UV variability observed at longer wavelengths with a characteristic time delay (Padovani et al. 2017), a *reverberation*, as discussed below.

Naturally, both propagating fluctuation lags and reverberation lags are simultaneously present in the data, but dominate at different temporal frequencies and have opposite energy dependences (Figure 2.1): at low frequencies ($\nu \lesssim 10^{-5}$ Hz for a $10^6 M_\odot$ black hole, or equivalently, $\gtrsim 1$ day), hard X-ray variations lag soft ones as accretion fluctuations propagate inward; at high frequencies ($\nu \gtrsim 10^{-3}$ Hz, timescales of minutes), soft X-rays lag hard ones as the disk responds to the driving corona (Kara and García 2025). Fourier analysis is essential for disentangling the two contributions.

Beyond this persistent stochastic variability, more extreme variability phenomena also exist, including *changing-look* AGN, which undergo dramatic changes in their broad emission lines and continuum on timescales of months to years, possibly driven by rapid changes in accretion rate or obscuration (LaMassa et al. 2015), as well as quasi-periodic eruptions (QPEs) and tidal disruption events (TDEs); these represent transient phenomena superimposed on the underlying stochastic variability and are not discussed further here.

¹The viscous timescale, $t_{\text{visc}} \sim r^2/\nu$, where ν is the kinematic viscosity, indicates the time for material to drift radially through the disk and increases with radius.

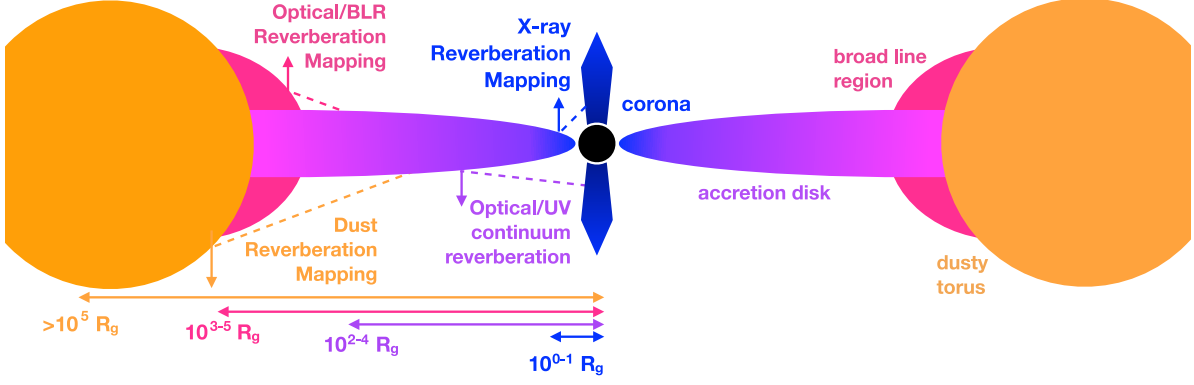


Figure 2.2: Cross-sectional schematic of an AGN depicting the four main types of reverberation mapping and their corresponding radial scales from the central black hole. Taken from Cackett et al. 2021.

2.1.1 Reverberation Mapping

The reverberation delays introduced above are not only a complication in the interpretation of light curves, they are also a powerful tool. Because the different regions of an AGN cannot be spatially resolved, their sizes cannot be measured directly, but the light-travel time between a driving source and a region that reprocesses its emission provides an indirect gauge. *Reverberation mapping* (RM) exploits this by monitoring the delay with which one part of the AGN responds to variations in another: a region at a distance R from the driving source responds on average after the light-crossing time $\tau_{\text{lag}} \sim R/c$, so that measuring the lag yields the size of the region (Cackett et al. 2021). The characteristic scales and their corresponding light-crossing times for the main AGN regions are summarised in Table 2.1, ranging from hours for the X-ray corona to years for the dusty torus.

Table 2.1: Order of magnitude size scales and their equivalent light-crossing times for the different regions of an AGN. Adapted from Cackett et al. 2021.

Region	Size scale (R_g)	Light-crossing time	
		$10^7 M_\odot$	$10^9 M_\odot$
X-ray corona	10	500 s	14 hr
UV/optical disk	10^2 – 10^4	0.06–6 days	6–600 days
BLR	10^3 – 10^5	0.6–60 days	60–6000 days
Dusty torus	$> 10^5$	> 60 days	> 6000 days

The two most established applications probe different regions. In *emission-line RM*, the broad emission lines respond to variations in the ionising continuum with a delay that measures the size of the BLR; combined with the velocity width of the lines, this yields an estimate of the black hole mass and underlies the radius-luminosity relation introduced in § 1.1.3 (Peterson 1993). In *continuum RM*, the delays are measured between different wavebands of the disk continuum itself, mapping the temperature structure of the accretion disk. For a standard thin disk the lag increases with wavelength as:

$$\tau_\lambda \propto \lambda^{4/3} \left[\frac{M_{\text{BH}} \dot{M}}{(3 + f_{\text{irr}})} \right]^{1/3}, \quad (2.1)$$

since longer wavelengths originate at larger, cooler radii, with the normalization set by the black hole mass, the accretion rate $\dot{M}$, and the fraction of irradiating flux f_{irr} (Cackett et al. 2021).

Formally, the observed response is described as the convolution of the driving light curve with a *transfer function* that encodes the geometry and kinematics of the responding region (Blandford and McKee 1982). Recovering this transfer function, or even the lag alone, is an ill-posed problem: the light curves are irregularly sampled and noisy, and different underlying signals can reproduce the same observations to within their uncertainties. This degeneracy is not unique to RM but is a general feature of inference from AGN light curves, and it motivates treating the reconstruction of a light curve probabilistically, returning a distribution of signals consistent with the data rather than a single best-fit curve. It is this probabilistic reconstruction of the individual light curve, rather than the measurement of lags between light curves, that is the subject of this thesis.

2.2 Variability Characterization

Variability has been studied under a variety of methods, starting from simple statistical quantities tracing the amplitude and timescales of the observed variability to more sophisticated approaches in both the time and frequency domains (Paolillo and Papadakis 2025). In general, these statistics attempt to measure the absolute or fractional variance of the light curve over some timescale Δt , or express the degree of correlation between different timescales or photometric bands (Ulrich et al. 1997; De Cicco et al. 2025).

The distribution of variability power over temporal frequencies (or, equivalently, 1/timescale) is quantified by the *power spectral density* (PSD), defined as the squared amplitude of the flux (e.g., Uttley et al. 2002). AGN PSDs typically follow a “red-noise” power law, $P(\nu) \propto \nu^{-\alpha}$ with $\alpha \sim 1$ at low frequencies, steepening to $\alpha \gtrsim 2$ above a characteristic break frequency ν_b , consequently displaying larger amplitude variations on longer timescales. This break timescale scales with black hole mass and accretion rate (McHardy et al. 2006), with smaller black holes varying more rapidly than bigger ones for the same observing duration (Figure 2.3). The anticorrelation can be interpreted as a consequence of the fact that the most luminous AGN are likely to host larger mass black holes (if all quasars accrete at similar Eddington rates), in which case, their variability should be diluted by the larger emitting region (Paolillo and Papadakis 2025).

The PSD also depends strongly on the observing waveband, with optical PSDs showing steeper slopes and lower normalization than X-ray PSDs, consistent with the propagating fluctuations picture where outer disk regions contribute power at lower frequencies and inner regions at higher ones (Kara and García 2025).

Both the amplitude and the timescale of the variability correlate with the physical properties of the accreting system. The *damping timescale* τ , which determines when the flux decorrelates from its past, increases with black hole mass and, more weakly, with rest-frame wavelength (e.g., Collier and Peterson 2001; Burke et al. 2021) linking the variability timescale and the thermal timescale of the accretion disk. The *variability amplitude*, in turn, decreases towards longer wavelengths, so that AGN vary with larger fractional amplitude in the bluer bands (MacLeod et al. 2010), consistent with the bluer emission arising from the hotter, more compact inner disk. These scaling relations, and in particular the dependence of τ on black hole mass, are used in Chapter 3 to assign physically motivated parameters to the simulated light curves.

A complementary characterization is provided in the time domain by the *structure function* (SF), which measures the mean variability amplitude as a function of the time lag Δt separating pairs of observations, adequate for the sparse, irregularly sampled light curves typical of optical

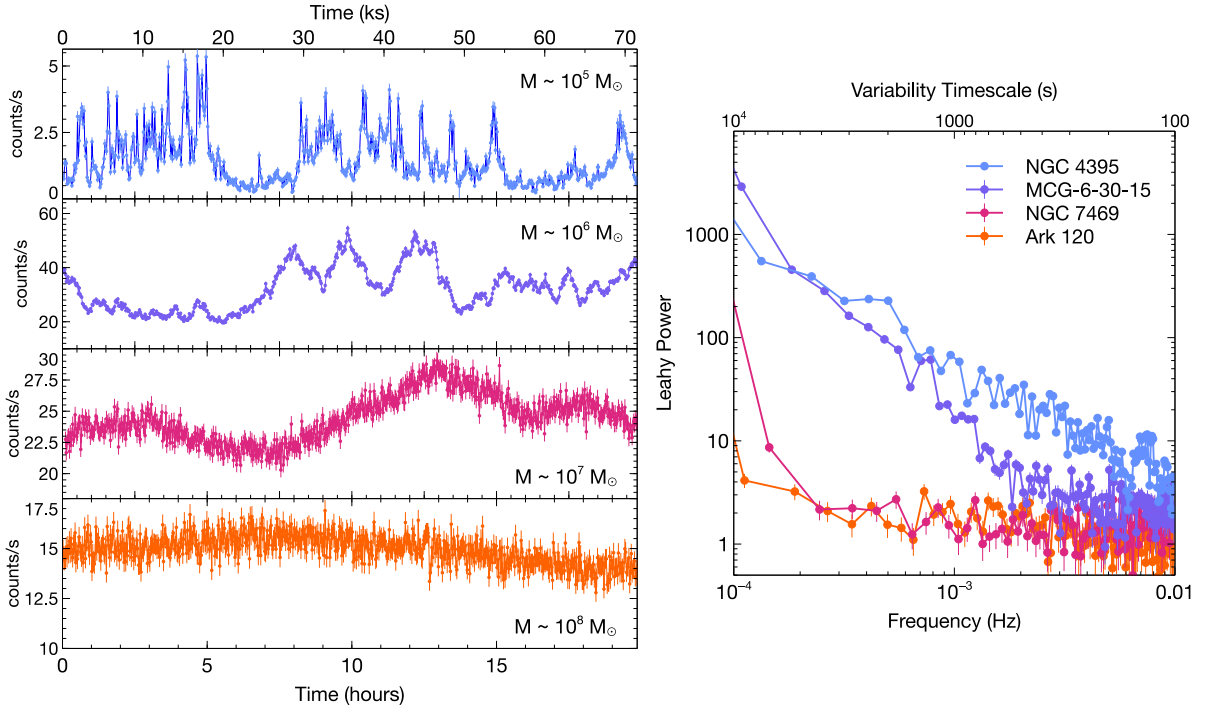


Figure 2.3: Left: X-ray light curves of four Seyfert 1 AGN with black hole masses spanning 10^5 – $10^8 M_\odot$, observed with XMM-Newton over 70 ks. Smaller black holes vary more rapidly, with the $10^5 M_\odot$ AGN (NGC 4395, top) showing variability on timescales of tens of seconds, while the $10^8 M_\odot$ AGN (Ark 120, bottom) varies on timescales of hours. Right: The corresponding PSDs, showing the red-noise power-law shape characteristic of AGN X-ray variability. Taken from Kara and García 2025.

surveys, where a direct estimate of the PSD is difficult (Kozłowski 2016b). On short lags the SF rises as a power law, reflecting the growing amplitude of the variability, and flattens above the damping timescale τ to a constant level set by the long-term variability amplitude, mirroring in the time domain the same turnover seen as the PSD break at $\nu_b \sim 1/\tau$. The shape of the SF therefore encodes the same two main quantities that characterize the variability as a whole, an amplitude and a timescale, which are precisely the parameters of the *damped random walk* (DRW) adopted to model the light curves in the following chapter.

3 Simulation Framework

The reconstruction method developed in this thesis is trained and tested on simulated light curves, for which the underlying signal is known exactly. This chapter sets out the framework used to generate them. It begins with the stochastic process and damped random walk, derives the transition probability that the reconstruction method will later learn, and then describes the simulation itself, including the way physically motivated timescales are assigned and the sampling regime.

3.1 Stochastic Processes

The variability introduced in Chapter 2 is aperiodic and aleatory, so it is not well described by a deterministic model with a fixed set of parameters. It is instead naturally treated as a realization of a *stochastic process*: a family of random variables indexed by time, whose statistical properties, rather than whose exact values, are the object of study. Rather than predicting the flux at a given epoch, one models the distribution from which that flux is drawn, conditioned on the values already observed.

The simplest stochastic model consistent with the observed variability of AGN is the *damped random walk* (DRW), also known as an Ornstein-Uhlenbeck process (Kelly, Bechtold, and Siemiginowska 2009). It is described by the stochastic differential equation

$$df(t) = -\frac{1}{\tau}f(t)dt + \sigma_d\sqrt{dt}\epsilon(t) \quad (3.1)$$

where τ is the damping timescale, σ_d sets the amplitude of the driving noise, and $\epsilon(t)$ is a white noise process¹. The two terms admit a direct physical reading. The first is a restoring term that pulls the flux back towards its mean on the timescale τ , so that excursions do not grow without bound, while the second perturbs it with stochastic kicks. The competition between damping and driving produces the characteristic behaviour of AGN light curves, which vary coherently on timescales shorter than τ and decorrelate into noise on timescales much longer than τ .

The DRW provides a good statistical description of optical AGN light curves (Kelly, Bechtold, and Siemiginowska 2009; Kozłowski et al. 2010; MacLeod et al. 2010) and, importantly, it is a *Gaussian process*, a distribution over functions such that the values at any finite set of times follow a multivariate normal distribution, specified by a mean function and a covariance function

¹A white noise process is one with a mean zero and no correlation between its values at different times.

(Rasmussen and Williams 2006). Conditioning on observed values yields a Gaussian predictive distribution in closed form, which is what makes such processes analytically tractable for interpolation. Its exponential autocorrelation function

$$\text{ACF}(\Delta t) = \exp\left(-\frac{|\Delta t|}{\tau}\right), \quad (3.2)$$

corresponding to a power spectrum of the form $P(\nu) \propto \nu^{-2}$ at high frequencies that flattens to white noise below the break frequency $\nu_b = 1/(2\pi\tau)$.

The damped random walk is also the simplest member of the broader family of *continuous-time autoregressive moving average* (CARMA) processes (Kelly et al. 2014), corresponding to the case CARMA(1, 0). A general CARMA(p , q) process is the solution of a linear stochastic differential equation of order p , driven by white noise through an order- q moving-average term, and can reproduce a wider range of power-spectrum shapes than the damped random walk alone. In particular the CARMA(2, 1) process, also known as the damped harmonic oscillator, has been found to describe some observed AGN light curves more faithfully (Yu et al. 2025). The damped random walk is nonetheless retained throughout this work as the generative model, both because it remains an adequate approximation on the timescales probed by the surveys considered here and because its analytic tractability provides a controlled setting in which to develop and test the reconstruction method. The consequences of this choice, in particular the effect of model misspecification when the method is applied to real data, are returned to in Chapter 5.

3.1.1 The Transition Probability

The exponential autocorrelation of Equation 3.2 has a consequence that is central to this work. Because the damped random walk is a Gaussian process, and because Eq. 3.1 is first-order and Markovian², the distribution of the state at a time $t + \Delta t$ given its value at time t is itself Gaussian and depends only on the elapsed time Δt . This *transition probability* is

$$X_{t+\Delta t} | X_t \sim \mathcal{N}\left(\mu + e^{-\Delta t/\tau}(X_t - \mu), \sigma^2(1 - e^{-2\Delta t/\tau})\right), \quad (3.3)$$

where μ is the long-term mean and σ the stationary standard deviation of the light curve. The mean relaxes exponentially from the current value towards μ on the timescale τ , while the variance grows from zero at $\Delta t = 0$ to the stationary value σ^2 as Δt becomes large compared with τ . Both limits are the expected ones. At short separations the process retains a strong memory of its previous state and the distribution is narrow and centred near X_t , whereas at separations much longer than τ the memory is lost and the distribution approaches the stationary marginal $\mathcal{N}(\mu, \sigma^2)$.

Equation 3.3 is the object that the reconstruction method of Chapter 4 learns from data. It provides both a target against which the learned transition can be validated and, through the same expression, the exact Gaussian process baseline against which the method is compared. After the light curves are centred, $\mu = 0$, the mean reduces to $X_t e^{-\Delta t/\tau}$ and the variance to $\sigma^2(1 - e^{-2\Delta t/\tau})$.

²A process is Markovian if its future depends only on its present state and not on the path by which it was reached, so that the state at one time contains all the information relevant to predicting later times.

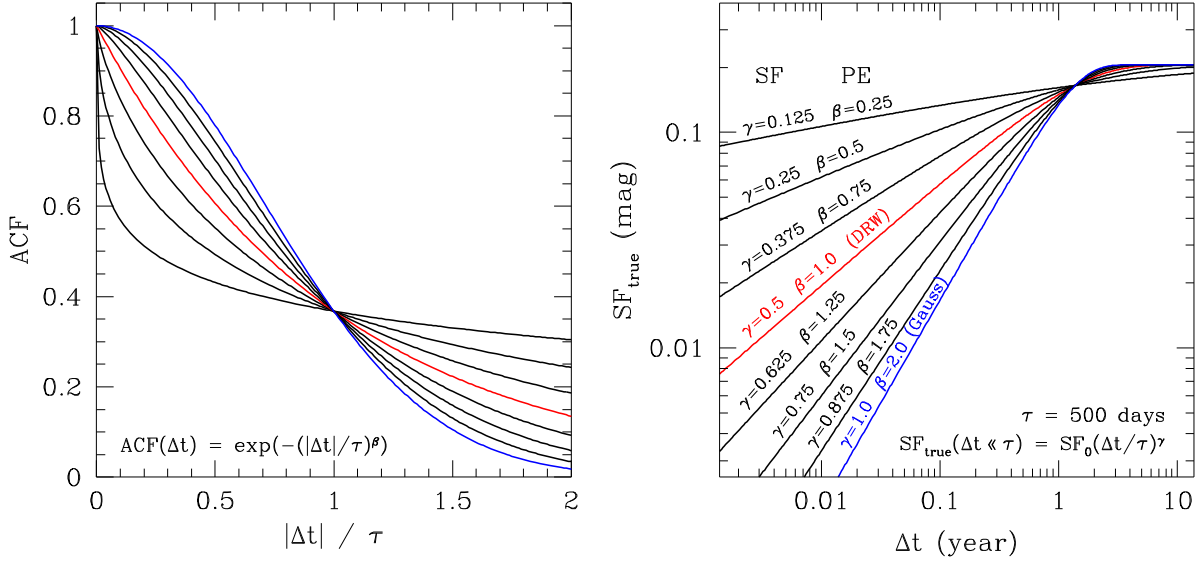


Figure 3.1: Left: autocorrelation functions of the family of Equation 3.4 for a range of the decay exponent β . Right: the corresponding structure functions, with short-timescale slope γ . The damped random walk ($\beta = 1$, red) produces a structure function of slope $\gamma = 0.5$ below the damping timescale, flattening above it, and sits between a slowly declining correlation and a Gaussian one ($\beta = 2$, blue). All curves cross at $\Delta t = \tau$, illustrating the role of the damping timescale as the characteristic decorrelation lag. Adapted from Kozłowski 2016b.

3.2 The Damping Timescale

The damping timescale is the quantity that most directly characterizes the variability, and it requires some care because its definition and its measured value depend on convention. In the structure-function representation, the damped random walk is one member of a family parametrized by the exponent of the autocorrelation decay,

$$\text{ACF}(\Delta t) = \exp\left(-\left(\frac{|\Delta t|}{\tau}\right)^\beta\right), \quad (3.4)$$

where $\beta = 1$ recovers the damped random walk and $\beta = 2$ a Gaussian correlation. The corresponding structure function rises as a power law on short timescales, $\text{SF}(\Delta t) \propto (\Delta t/\tau)^\gamma$ for $\Delta t \ll \tau$, before flattening above the damping timescale; the damped random walk corresponds to a short-timescale slope $\gamma = 0.5$ (Figure 3.1). Real light curves are consistent with exponents scattered around these values, which is one of the empirical justifications for adopting the damped random walk as a baseline while remaining aware of its limitations.

A second source of ambiguity is the reference frame. The damping timescale measured from an observed light curve is stretched relative to its rest-frame value by a factor of $(1 + z)$, so that comparisons between sources at different redshifts, or between measurements and physical models, must specify whether τ is quoted in the observed or the rest frame. Throughout this work τ refers to the rest-frame timescale unless stated otherwise.

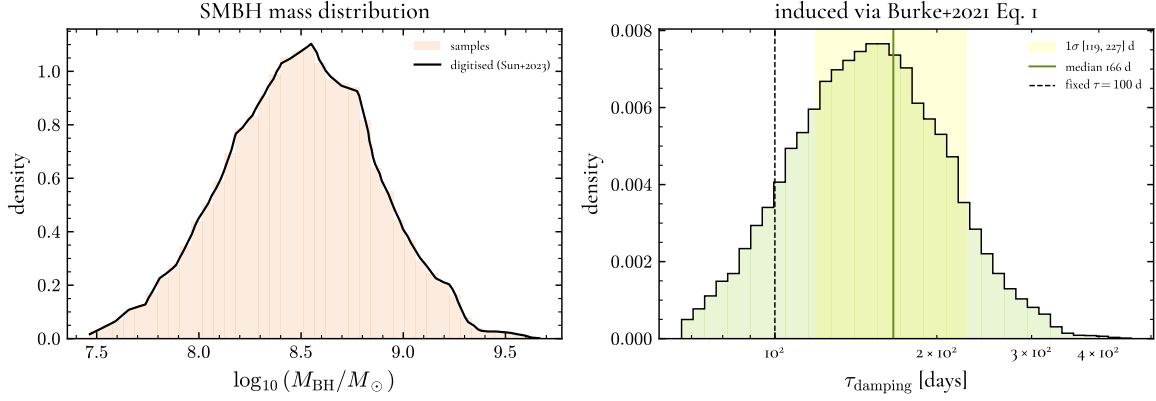


Figure 3.2: Left: the black hole mass distribution used to assign timescales, digitized from Sun 2023, with samples drawn from it by inverse-transform sampling. Right: the damping-timescale distribution induced by passing these masses through the mass-timescale relation of Equation 3.5, with median 166 d and the marked 1σ range [119, 227] d. The fixed value $\tau = 100$ d used in the controlled experiment is shown for reference.

3.3 Physically Motivated Timescales

For the simulated light curves to resemble a real population, the damping timescale is not fixed at a single value but drawn from a distribution representative of observed AGN. As introduced in Chapter 2, τ increases with black hole mass, therefore timescale is assigned in two steps: first, a black hole mass is first drawn from the bias-corrected mass distribution of quasars of Sun (2023), and the corresponding damping timescale is then obtained from the mass-timescale relation of Burke et al. (2021),

$$\tau = 107^{+11}_{-12} \text{ days} \left(\frac{M_{\text{BH}}}{10^8 M_{\odot}} \right)^{0.38^{+0.05}_{-0.04}}. \quad (3.5)$$

The mass distribution is digitized from the observed sample and sampled in full by inverse-transform sampling, so the resulting timescales inherit its shape through Equation 3.5, including its high-mass tail. The induced distribution has a median of $\tau \approx 166$ d, with the central 68% of curves falling between 119 and 227 d (Figure 3.2).

This procedure defines the varying-timescale configuration used for most of the experiments, in which each light curve is assigned its own τ and the reconstruction method must generalize across the range shown in Figure 3.2. A simpler configuration was also used, in which the timescale was held fixed at $\tau = 100$ d for all curves and served as a controlled setting in which the sampling and noise can be studied in isolation from any variation in τ . The varying-timescale case is the more realistic and more demanding test, and is the one against which the main results are reported.

3.4 Light Curve Generation

Light curves are generated with the EZTAO package (Yu and Richards 2022), which represents the damped random walk as a Gaussian process and evaluates it through the *celerite* method (Foreman-Mackey et al. 2017). A direct evaluation of the Gaussian process likelihood requires inverting the covariance matrix at a cost that scales as $\mathcal{O}(N^3)$ in the number of epochs, which is prohibitive

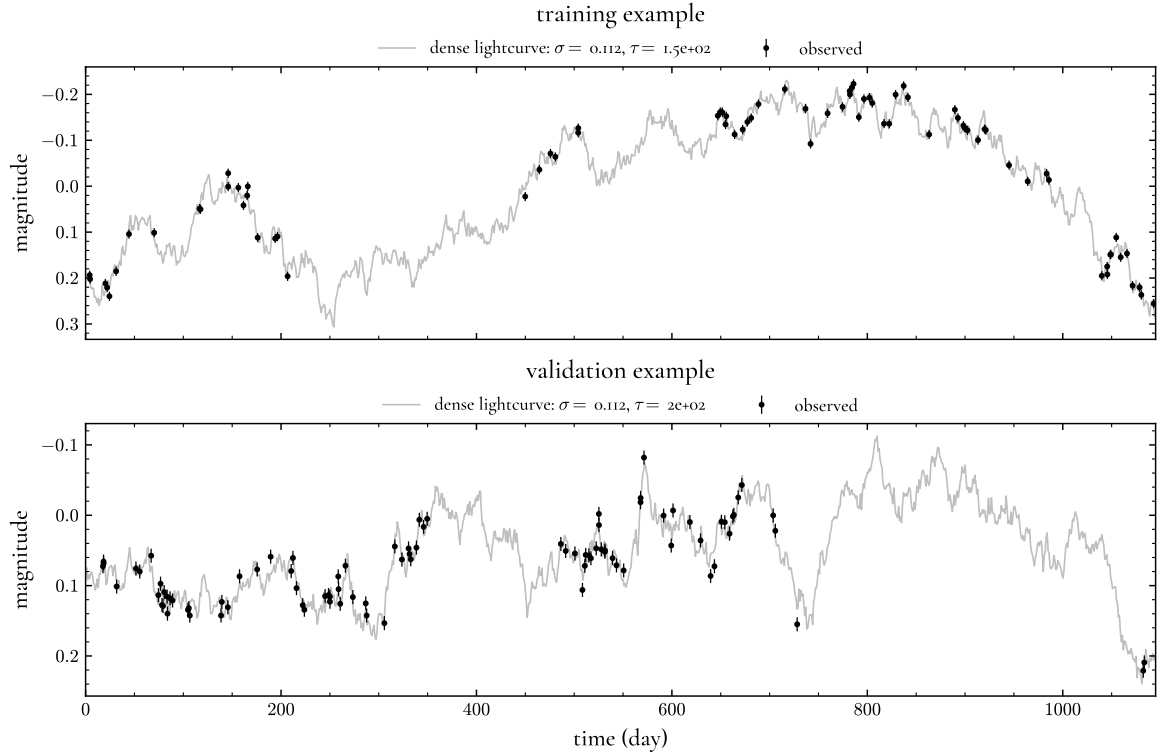


Figure 3.3: Two simulated light curves (one from the training set and one from the validation set), each showing the dense underlying realization (grey) and the sparse, noisy observations drawn from it (black).

for long light curves; celerite reduces this to $\mathcal{O}(N)$ for the damped random walk kernel, making the simulation of large samples tractable.. Modeling is carried out on magnitudes rather than fluxes, since the flux light curves of accreting sources are approximately lognormal and so are close to Gaussian in magnitude, matching better the assumption of the Gaussian process. For each simulated object a damping timescale is assigned as described above, the stationary amplitude is set to $\sigma = 0.112$ mag, a value representative of quasar optical variability taken from Yu et al. (2025), and a densely sampled realization is produced to represent the true, underlying signal.

The observed light curve is then formed from this dense realization rather than interpolated from it. An irregular cadence is drawn, the dense realization is evaluated on the union of the true grid and the observed epochs in a single simulation, and the observed magnitudes are read off at the sampled times and perturbed with photometric noise. This construction ensures that the observations and the underlying signal are two views of one realization, and gives access to the true signal at every epoch, which is what allows the reconstruction to be evaluated exactly. The photometric noise is added as a Gaussian perturbation at each epoch; for the simulations a single-visit uncertainty of 0.01 mag is adopted, matching the expected photometric precision of the Legacy Survey of Space and Time (LSST) at the Vera C. Rubin Observatory³. An example of each of the two configurations is shown in Figure 3.3.

³<https://rubinobservatory.org/for-scientists/rubin-101/key-numbers>

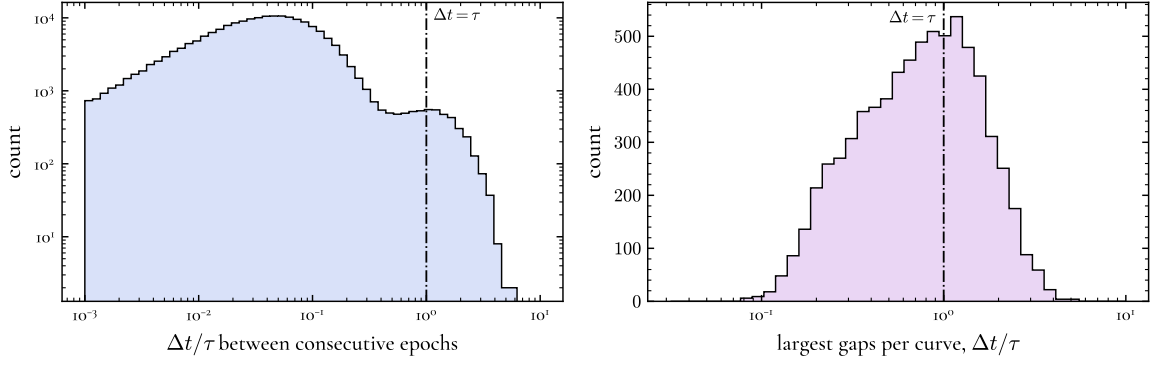


Figure 3.4: Left: the distribution of the spacing between consecutive epochs, in units of the damping timescale. Right: the distribution of the single largest gap in each light curve, in the same units. Most consecutive spacings lie well below τ , while the largest gaps straddle τ , the regime in which reconstruction is most demanding. The line marks $\Delta t = \tau$.

3.5 Sampling Regime

The difficulty of gap filling is controlled by a single dimensionless quantity, the gap length in units of the damping timescale, $\Delta t/\tau$, because a damped random walk loses memory as $e^{-\Delta t/\tau}$. The sampling parameters are therefore quoted in units of τ rather than in days. With a three-year baseline ($t_{\max} = 1095$ d), 130 attempted epochs, and a gap fraction of 0.35 distributed over three windows, at the fiducial timescale $\tau = 100$ d the baseline spans 10.9τ , the mean spacing between consecutive epochs is 8.4 d $= 0.084\tau$, and the mean gap is 128 d $= 1.28\tau$. Consecutive epochs are therefore strongly correlated, while the gaps straddle the decorrelation scale, the regime in which reconstruction is neither trivial nor impossible. Figure 3.4 shows the resulting distributions of the consecutive spacing and of the largest gap in each curve, both in units of τ .

This regime is bounded by two thresholds, both of which appear in the literature as biases on the recovery of τ . The timescale is underestimated when the baseline is shorter than roughly 10τ (Kozłowski 2016a), and overestimated when it is shorter than about five times the mean cadence (Hu 2024). Here they matter for a different reason: they delimit the range over which a gap carries information at all. Below the cadence threshold consecutive epochs are effectively uncorrelated and no interpolation is possible, so a model can only return the prior; above the baseline threshold the signal barely decorrelates over the observing window and every gap is trivially interpolable. This is also why the fiducial experiments fix $\tau = 100$ d rather than adopting a broad range, since most of a wide range would produce degenerate light curves at this baseline and cadence. Where a spread of timescales is introduced, the physically motivated distribution of Section 3.3, with its bulk between roughly 100 and 230 d, lies within the informative range.

A single photometric band is used throughout. Although surveys such as ZTF and LSST observe in several bands, and although the coordinated multi-band variability discussed in Chapter 2 is in principle a rich source of additional information, the treatment here is confined to single-band light curves, and the extension to multiple bands is left as a direction for future work.

4 Neural Stochastic Flows

The reconstruction of a gappy light curve was framed in § 2.1.1 as an ill-posed problem, best approached by returning a distribution of signals consistent with the data rather than a single curve. The damped random walk introduced in Chapter 3 already provides such a probabilistic description because it is a Gaussian process and the transition between any two epochs has a known analytic form. The method adopted in this thesis replaces that transition with a flexible one learned directly from data, using *neural stochastic flows* (NSF; Kiyohara et al. 2025). The motivation is twofold, flexibility and efficiency. A learned transition can in principle capture departures from the damped random walk that a fixed kernel cannot, and, more generally, neural stochastic flows are *solver-free*: they model the transition density of a stochastic differential equation directly, without the numerical integration that conventional neural SDEs require. In the damped random walk case considered here the transition is analytically known, which provides an exact reference against which the learned model can be validated (see Chapter 5).

4.1 Normalizing Flows

A *normalizing flow* (NF) transforms a simple probability distribution, typically a standard normal $z \sim p_Z(z)$, into a more complex one through a sequence of invertible and differentiable mappings (Rezende et al. 2016; Kobyzev et al. 2021). A sample is generated by passing the base variable through the map, $y = g_\theta(z)$, the *generative direction*. Because the map is invertible, one can also go back from y to z through $z = f_\theta(y) = g_\theta^{-1}(y)$, the *normalizing direction*, and the density of y follows from the change-of-variables formula,

$$p_Y(y) = p_Z(f_\theta(y)) \left| \det J_{f_\theta}(y) \right|, \quad (4.1)$$

where J_{f_θ} is the Jacobian of the inverse map. The base density p_Z is known and the Jacobian term, the *volume correction*, accounts for the change of volume induced by the transformation, so Eq. 4.1 yields an exact density for y (Figure 4.1). This exactness, together with the ability both to sample and to evaluate densities, is what makes normalizing flows well suited to modelling complex distributions, and it means the model can be trained simply by maximising the likelihood of the data under Eq. 4.1 (Kobyzev et al. 2021).

For Eq. 4.1 to be practical, the transformation must satisfy two requirements: it must be easily invertible, and its Jacobian determinant must be cheap to compute (Kobyzev et al. 2021). A construction that meets both, and the one on which the model used here is built, is the *coupling flow*

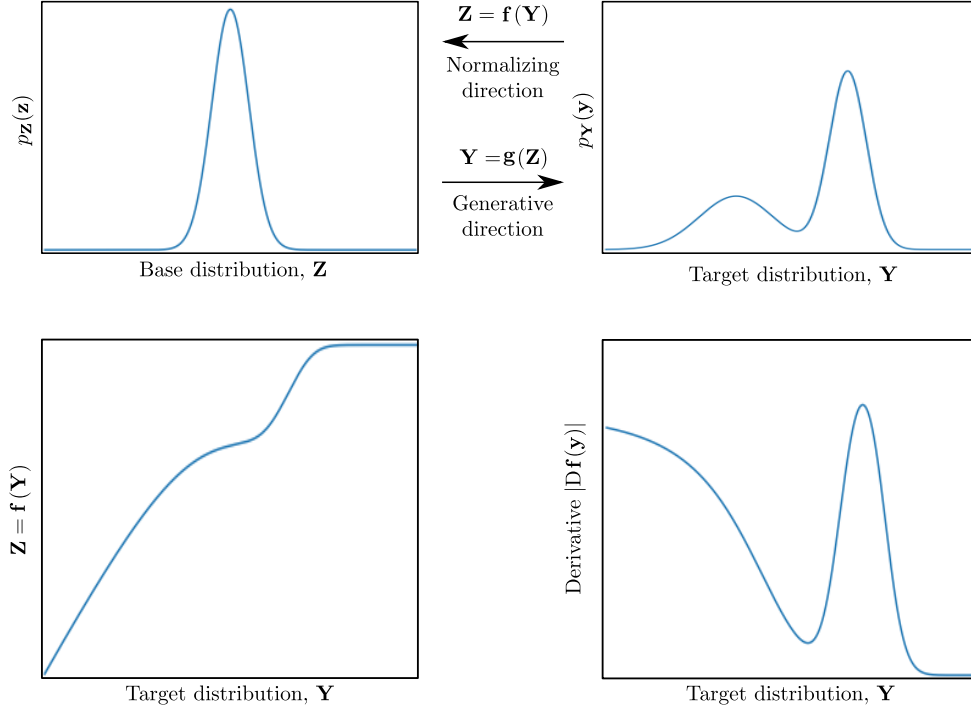


Figure 4.1: The change of variables of Eq. 4.1. Top left: the base density p_Z . Top right: the target density p_Y . A bijective map g transforms one into the other, with inverse f . Bottom left: the inverse map f . Bottom right: its absolute Jacobian, which rescales the density so that probability is conserved. Adapted from Kobyzev et al. 2021.

(Dinh, Krueger, and Bengio 2015). The input is split into two partitions; one is left unchanged, and the other is transformed by a bijection whose parameters are produced by a *conditioner* network taking the unchanged partition as input,

$$y_A = x_A, \quad y_B = x_B \odot \exp(s(x_A)) + t(x_A), \quad (4.2)$$

where s and t are the scale and shift functions and $\odot$ is the elementwise product. This is the affine coupling layer of RealNVP (Dinh et al. 2017). It is trivially inverted, since the unchanged partition $x_A = y_A$ recovers the arguments of s and t , and its Jacobian is triangular, so its determinant is simply the product of the diagonal, $\prod_j \exp(s(x_A))_j$. Because neither the inverse nor the determinant requires inverting or differentiating s and t , these functions can be arbitrarily complex neural networks. A single such layer leaves one partition untouched, so the partitions are alternated between successive layers, and stacking many layers yields an expressive invertible map.

4.2 From Static Densities to Transition Laws

Neural stochastic flows extend this construction from a single *static* density to the *transition law* of a stochastic process (Kiyohara et al. 2025). Rather than modelling one distribution $p_Y(y)$, they model the conditional distribution of a future state given a current state and an elapsed time,

$$p_\theta(x_t \mid x_s, \Delta t), \quad \Delta t = t - s, \quad (4.3)$$

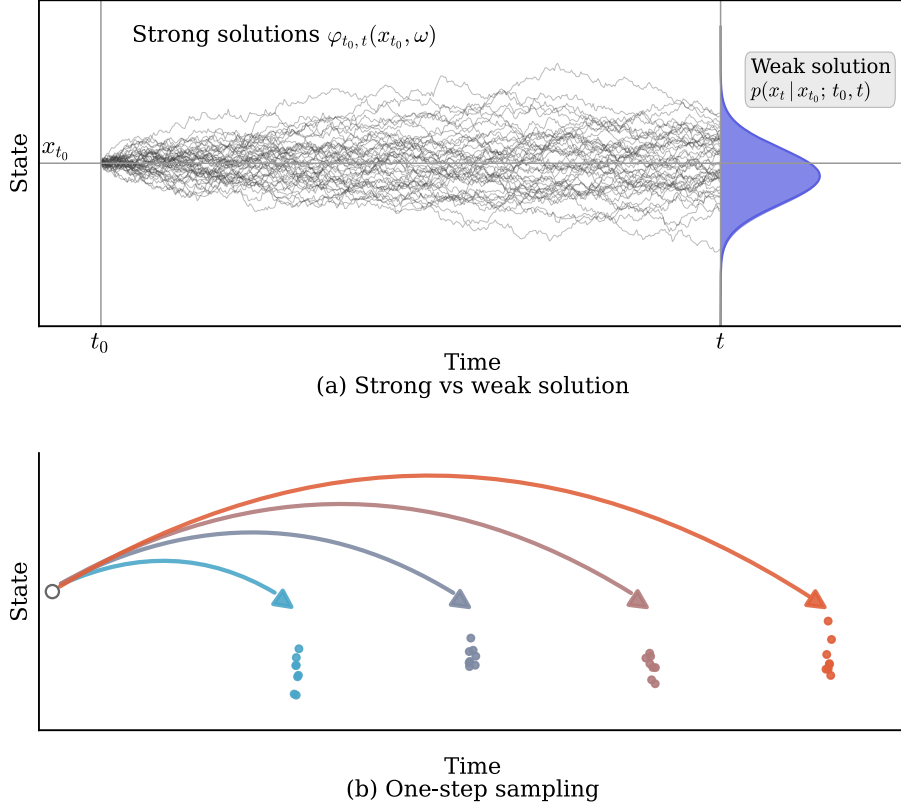


Figure 4.2: (a) The *strong solution* of a stochastic differential equation is an ensemble of paths $\varphi_{t_0,t}(x_{t_0}, \omega)$ diverging from a common initial state; the *weak solution* is the transition density $p(x_t | x_{t_0}; t_0, t)$ they define at a later time, shown at the right edge. Neural stochastic flows model this density directly. (b) Because the density is modelled directly, samples at any elapsed time can be drawn in a single step, without the iterative integration that path-based methods require. Adapted from Kiyohara et al. 2025.

that is, the distribution of the state x_t at time t given that the process was at x_s at the earlier time s . This conditional is defined in the same way as an ordinary flow, but with the transformation now conditioned on the starting state and the elapsed time: a Gaussian noise variable $\epsilon \sim \mathcal{N}(0, I)$ is passed through a conditional normalizing flow,

$$x_t = F_\theta(x_s, \Delta t, \epsilon). \quad (4.4)$$

For fixed x_s and Δt , different draws of ϵ produce different possible future states x_t , and together they define the conditional density $p_\theta(x_t | x_s, \Delta t)$ of Eq. 4.3. Modelling this density directly, rather than the individual paths of the process, is what allows a state at any elapsed time to be sampled in a single step, without iterative integration (Figure 4.2).

Concretely, the sample is drawn by first initializing a state-dependent Gaussian that mirrors an Euler-Maruyama step, with a mean displaced from x_s by a drift scaled by Δt and a variance scaled by $\sqrt{\Delta t}$,

$$z = \underbrace{x_s + \Delta t \text{MLP}_\mu(c)}_{\mu(c)} + \underbrace{\sqrt{\Delta t} \text{MLP}_\sigma(c)}_{\sigma(c)} \odot \epsilon, \quad c := (x_s, \Delta t), \quad (4.5)$$

and then passing it through a sequence of coupling layers of the form of Eq. 4.2, whose scale and shift functions are now additionally conditioned on c and multiplied by Δt . The map F_θ

is constrained in this way so that the transition density inherits the defining properties of the solution of a stochastic differential equation (Kiyohara et al. 2025). Two are relevant here. The *identity* property requires that a zero time gap returns the input state, $p_\theta(x_t | x_s, 0) = \delta(x_t - x_s)$; the explicit factors of Δt in Equations 4.5 and 4.2 enforce this, since both the Gaussian and every coupling layer collapse to the identity as $\Delta t \rightarrow 0$. The *stationarity* property requires that, for a time-homogeneous process, the transition depend only on the gap Δt and not on absolute time; since the damped random walk is autonomous, the conditioner omits absolute time and depends on Δt alone. A third property, discussed in the following section, ties transitions of different lengths together.

This learned transition plays the role of the damped random walk step in place of the analytic transition of the Ornstein-Uhlenbeck process (Chapter 3), the flexible map F_θ models the variability between any two epochs, while reducing to the correct transition when the data are generated by a damped random walk.

4.3 Triplets and Stochastic Bridges

Reconstructing a gap requires the distribution of a state that lies *between* two observations, not merely one that follows from a single earlier state. This intermediate conditional is modelled by a *stochastic bridge*. Following the training strategy of Kiyohara et al. (2025), from each light curve one randomly samples triplets of times (t_i, t_j, t_k) with $t_i < t_j < t_k$, and two models are trained jointly on them. The *flow* model learns the transitions between states, both the direct endpoint transition $p_\theta(x_k | x_i, \Delta t_{ik})$ and the two one-step transitions $p_\theta(x_j | x_i, \Delta t_{ij})$ and $p_\theta(x_k | x_j, \Delta t_{jk})$. The *bridge* model learns the intermediate conditional¹

$$q_\xi(x_j | x_i, x_k, t_i, t_j, t_k), \quad (4.6)$$

the distribution of the middle state given both surrounding states and their times. It is this bridge, conditioned on the two bracketing observations, that performs the gap filling at inference.

The two models are tied together by the third inherited property, the *flow property*, which requires that the transition density satisfy the Chapman-Kolmogorov relation: the direct transition from x_i to x_k must equal the composition of the two one-step transitions through any intermediate state,

$$p_\theta(x_k | x_i, \Delta t_{ik}) = \int p_\theta(x_k | x_j, \Delta t_{jk}) p_\theta(x_j | x_i, \Delta t_{ij}) dx_j. \quad (4.7)$$

Equivalently, the joint distribution of the pair (x_j, x_k) given x_i can be factorized either as a one-step flow from x_i to x_k followed by the bridge that places x_j between them, or as a two-step flow $x_i \rightarrow x_j \rightarrow x_k$; both describe the same distribution and must agree. Enforcing this relation is what forces the flow and the bridge to represent a single, self-consistent stochastic process rather than two unrelated conditionals.

The two sides of Eq. 4.7 cannot be compared directly, since the two-step side requires marginalizing over the intermediate state x_j . The bridge q_ξ is therefore introduced as an auxiliary variational distribution, which makes the comparison tractable through variational upper bounds on the Kullback-Leibler divergence between the one-step and two-step transitions, taken in both

¹The bridge is denoted b_ξ in the original formulation of Kiyohara et al. (2025); it is written q_ξ here to emphasize its role as an auxiliary variational distribution and for consistency with the notation used throughout this work.

directions since the divergence is asymmetric. The forward bound, from the one-step to the two-step transition, is

$$\mathcal{L}_{1\text{-to-}2} = \mathbb{E} \left[\log \frac{p_{\theta}(x_k | x_i) q_{\xi}(x_j | x_i, x_k)}{p_{\theta}(x_k | x_j) p_{\theta}(x_j | x_i)} \right], \quad (4.8)$$

with the expectation over $p_{\theta}(x_k | x_i) q_{\xi}(x_j | x_i, x_k)$, and the reverse bound is

$$\mathcal{L}_{2\text{-to-}1} = \mathbb{E} \left[\log \frac{p_{\theta}(x_j | x_i) p_{\theta}(x_k | x_j)}{q_{\xi}(x_j | x_i, x_k) p_{\theta}(x_k | x_i)} \right], \quad (4.9)$$

with the expectation over $p_{\theta}(x_j | x_i) p_{\theta}(x_k | x_j)$. Their sum is the flow-consistency loss, $\mathcal{L}_{\text{flow}} = \mathcal{L}_{1\text{-to-}2} + \mathcal{L}_{2\text{-to-}1}$.

4.4 Training Objective

The model is trained by minimising a loss with two terms (Kiyohara et al. 2025),

$$\mathcal{L}(\theta, \xi) = \mathcal{L}_{\text{data}}(\theta) + \lambda \mathcal{L}_{\text{flow}}(\theta, \xi), \quad (4.10)$$

where $\mathcal{L}_{\text{data}}$ is the negative log-likelihood of the observed transitions under the flow, evaluated through the change-of-variables formula of Eq. 4.1, and $\mathcal{L}_{\text{flow}}$ is the flow-consistency term of Equations 4.8 and 4.9, weighted by a hyperparameter λ . The data term anchors the model to the observed light curves, while the consistency term enforces that the learned transitions compose correctly across time. Both terms are necessary: the consistency term alone admits self-consistent but arbitrary transitions, and it is the data likelihood that ties the learned process to the damped random walk actually present in the light curves. The sensitivity of the results to the weighting λ is examined in Chapter 5.

4.5 Implementation

The model is implemented in JAX (Bradbury et al. 2018), building on the publicly available `jax_nsf` codebase accompanying Kiyohara et al. (2025), from which the flow and bridge architectures and the core training machinery are taken. The change-of-variables likelihood of Eq. 4.1 and the variational bounds of Equations 4.8 and 4.9 are differentiated automatically, so the gradients of the loss are obtained straightforward, and the full training step is compiled once and reused across iterations, which makes the repeated evaluation of the loss over many triplets efficient. The flow and bridge are represented as `equinox` modules (Kidger and Garcia 2021), and only their floating-point array leaves are treated as trainable parameters, with the remaining static structure held fixed during optimisation.

Training minimizes the objective of Eq. 4.10 using the Adam optimizer (Kingma and Ba 2017) as implemented in `optax` (DeepMind et al. 2020), with a learning rate of 2×10^{-3} . At each step a batch of sixteen light curves is drawn, and from each a set of thirty-two triplets (t_i, t_j, t_k) is sampled, with the intermediate times drawn log-uniformly so that both short and long gaps are represented. The loss is averaged over the batch and its gradient taken through the compiled step. The zero-padding introduced when the ragged light curves are stored as fixed-size arrays (Chapter 3) is excluded from the loss through a mask, so that only genuine observations contribute.

The simulation of the light curves and the training of the model are carried out in two separate environments. The simulator relies on **EzTao** (Yu and Richards 2022), whose dependencies conflict with those of **jax_nsf**, so the two cannot share an environment. The simulated light curves are written to disk and read back for training, with the saved dataset acting as the only interface between the two stages. This separation keeps the generation of the training set reproducible and independent of the model, so a dataset can be regenerated or reused without retraining.

The contributions of this thesis are in the surrounding framework rather than in the flow architecture itself: the simulation of damped random walk light curves with physically motivated parameters (Chapter 3), the construction of a training set matched to the sampling and noise of a real survey (Chapter 5), and the evaluation of the learned model against the analytic transition and a Gaussian process baseline.

5 Results and Discussion

5.1 Fixed Characteristic Timescale τ

5.2 Sampling timescales

There is broader regimes of the τ that were not explored

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